Imperial College London

The Science of Climate Change

Climate feedbacks & sensitivity

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Today's topics

- The energy budget equation
- Climate sensitivity
- Transient climate response
- Climate feedbacks

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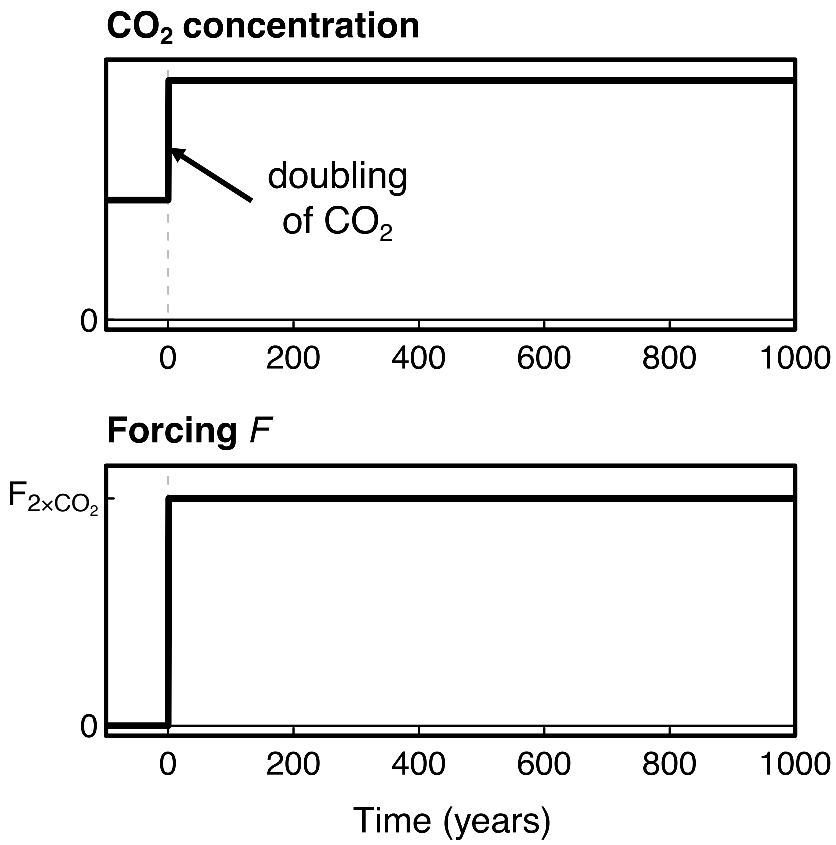
Perturbing the radiative balance

- Consider the following scenario: CO_2 concentration suddenly doubles, causing a positive forcing F_{2xCO2}
- The forcing then remains constant for many centuries



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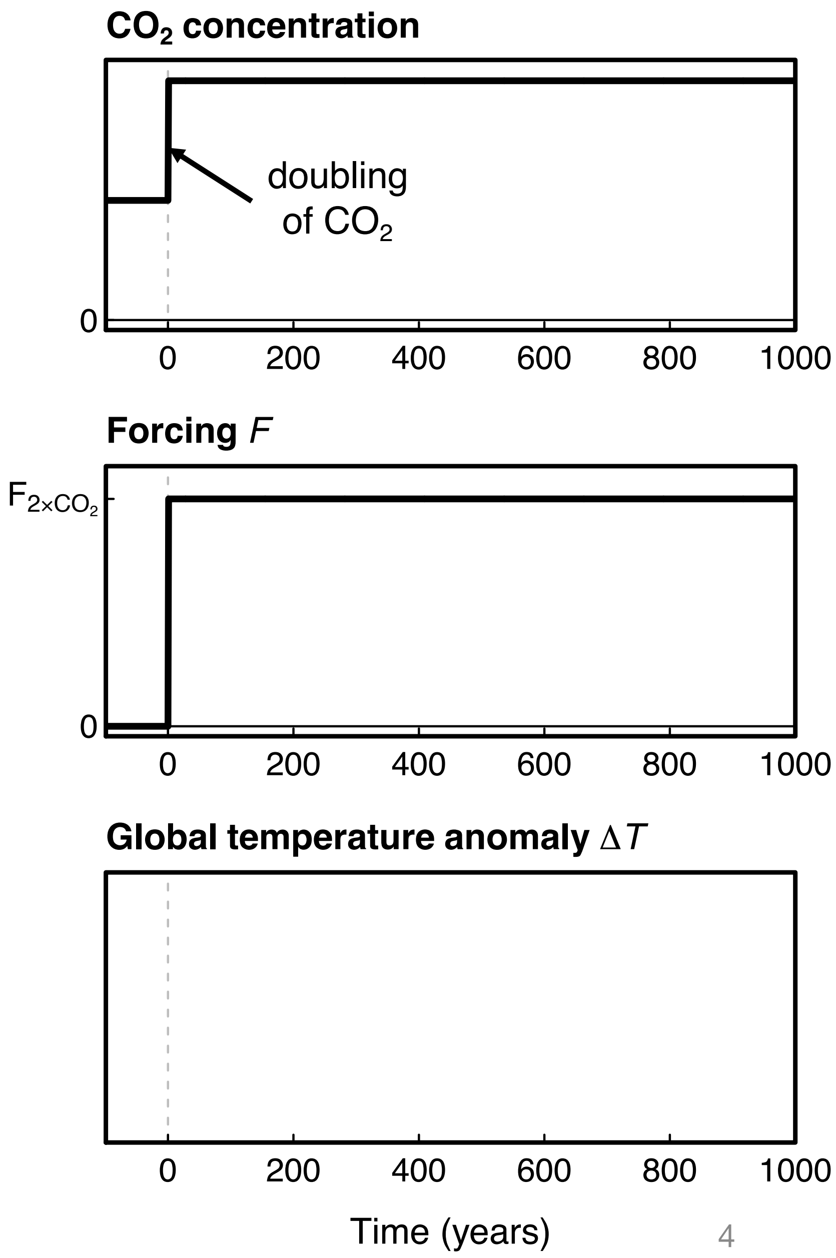

Perturbing the radiative balance

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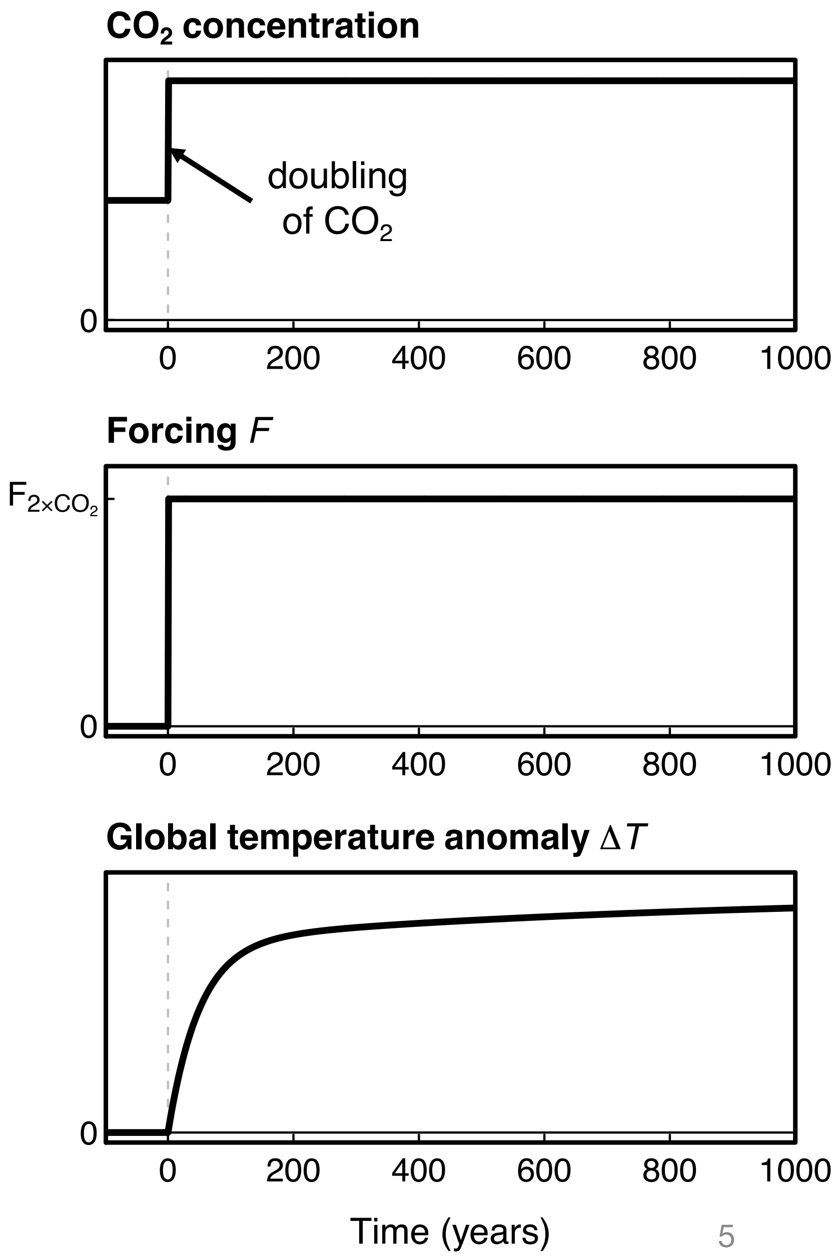
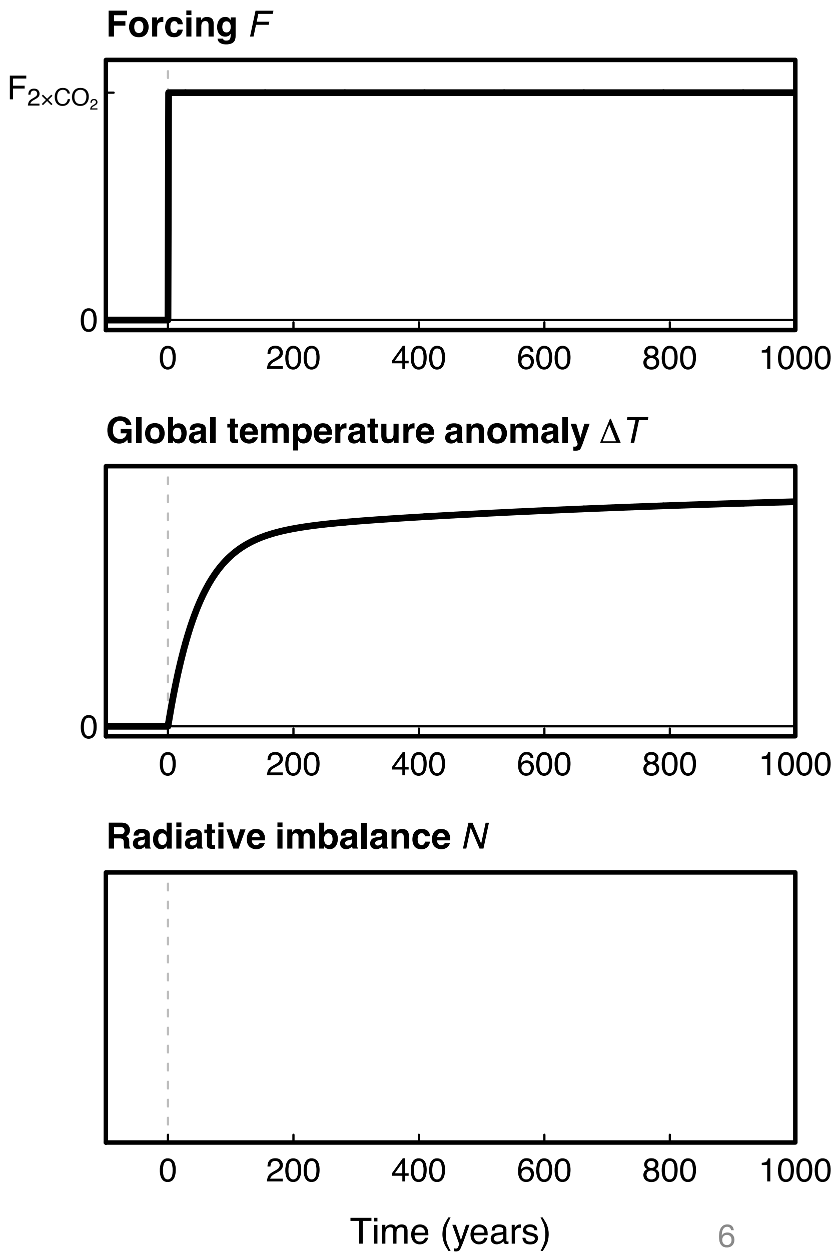
- The forcing then remains constant for many centuries
- How will global-mean temperature evolve?
- It will warm gradually over many centuries



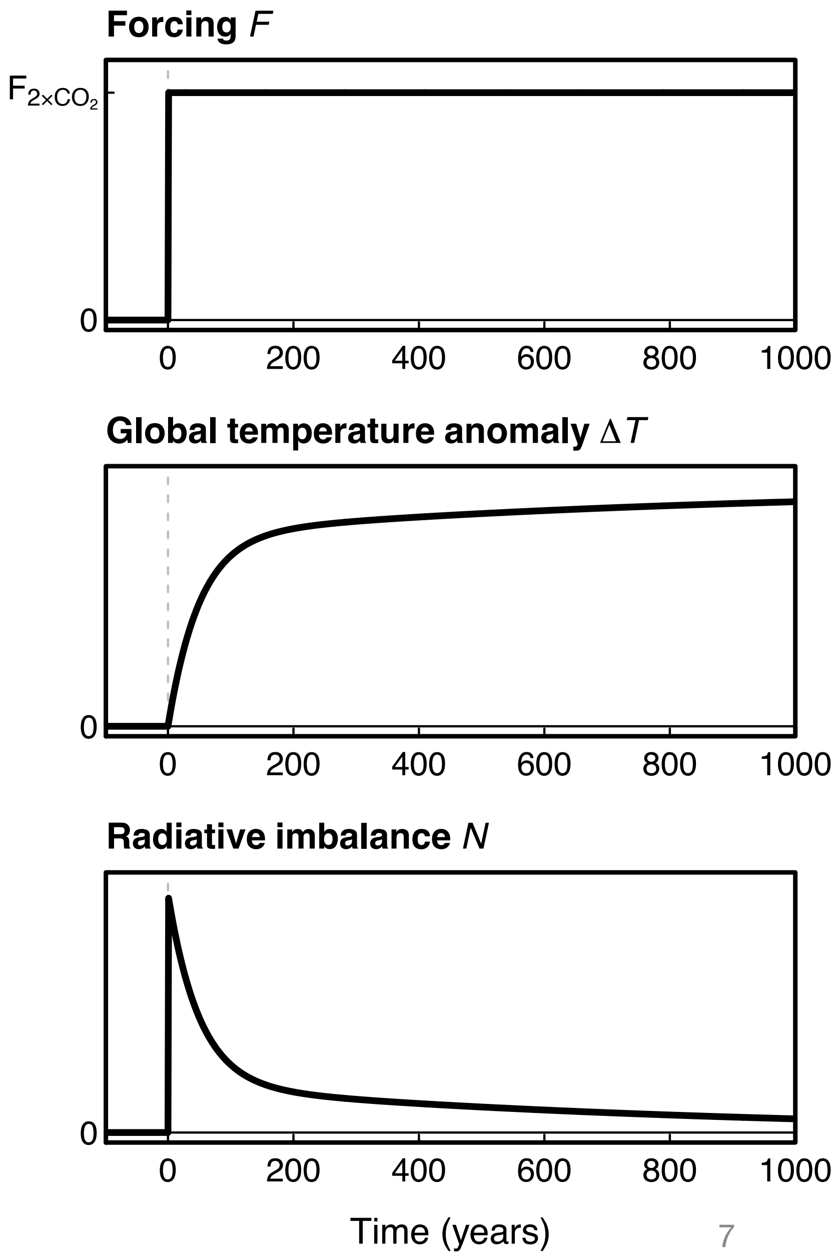
Perturbing the radiative balance

- Let's now define the **radiative imbalance** N = "in" minus "out"
- How does N evolve before and after the forcing?



Perturbing the radiative balance

- Let's now define the **radiative imbalance** N = "in" minus "out"
- How does N evolve before and after the forcing?
- It rises with the forcing: at time $t = 0$, $N = F_{2xCO2}$ (by definition)
- N then gradually returns to 0 as the Earth warms, since warming gradually increases emission to space

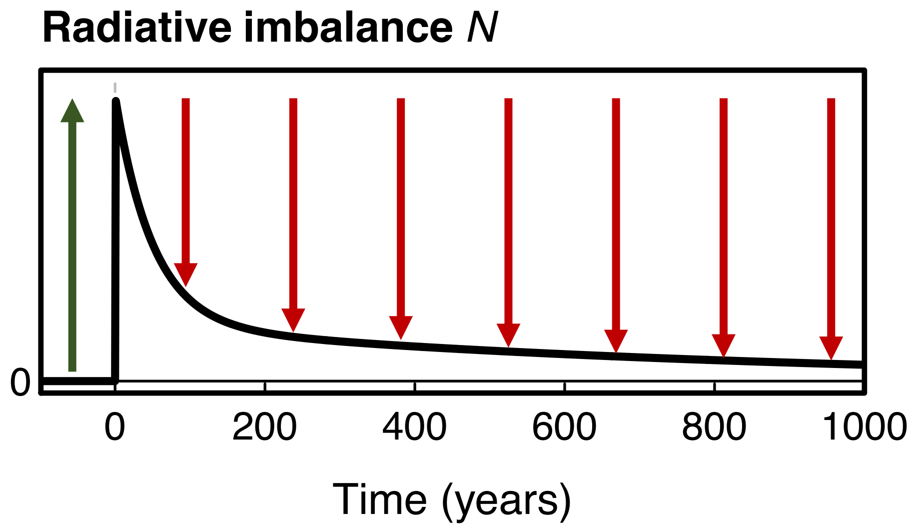


The energy budget equation

- We can then write the **energy budget equation** as

$$N = F - \lambda \Delta T$$

- ***F*: radiative forcing**
- ***λΔT*: radiative response**, opposing the forcing. Scales with the global temperature anomaly, ΔT



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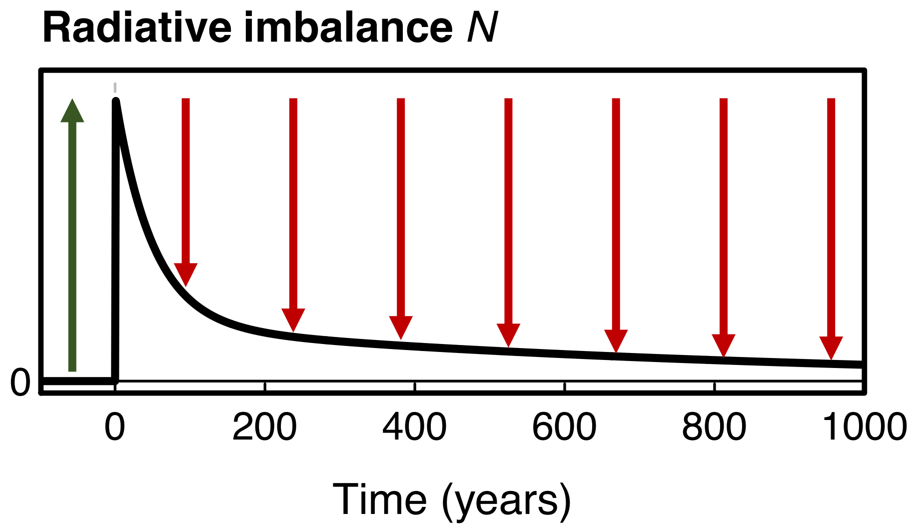
- ***F*: radiative forcing**
- ***λΔT*: radiative response**, opposing the forcing. Scales with the global temperature anomaly, ΔT
- If $F > 0$, then the Earth needs to warm in order to emit more infrared energy to space and re-establish radiative balance. Vice versa if $F < 0$.
- λ ($W/m^2/°C$) determines how much the outflow of energy to space changes per unit global warming.



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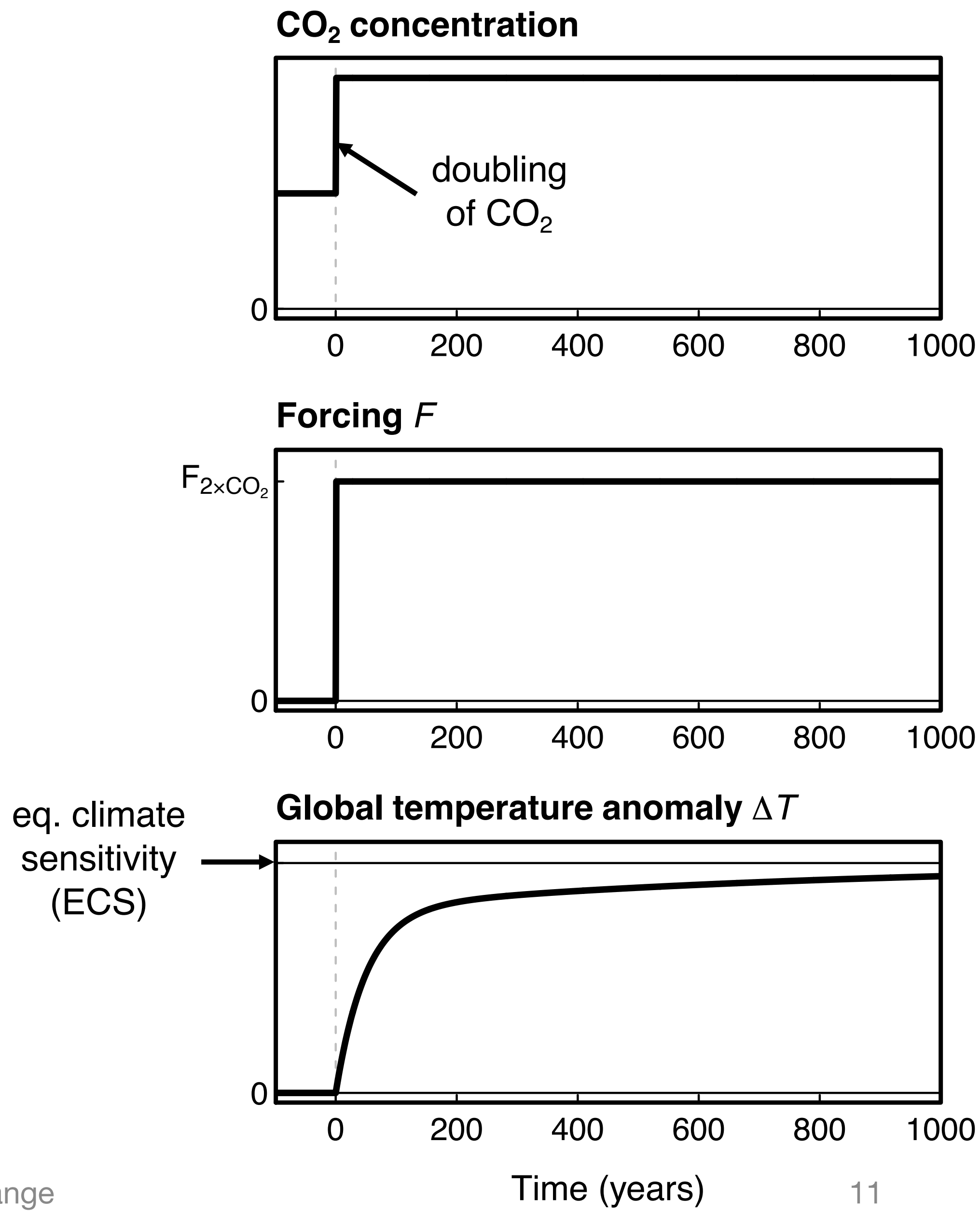
Climate sensitivity

- (Equilibrium) climate sensitivity is the amount of global warming, **at equilibrium**, after a doubling of CO_2
- In practice, it would take a very long time (>1000 years) to achieve equilibrium after doubling CO_2
- This is because the oceans warm very slowly (high heat capacity and slow mixing rate)



Climate sensitivity

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$$0 = F_{2xCO2} - \lambda \Delta T_{2xCO2}$$

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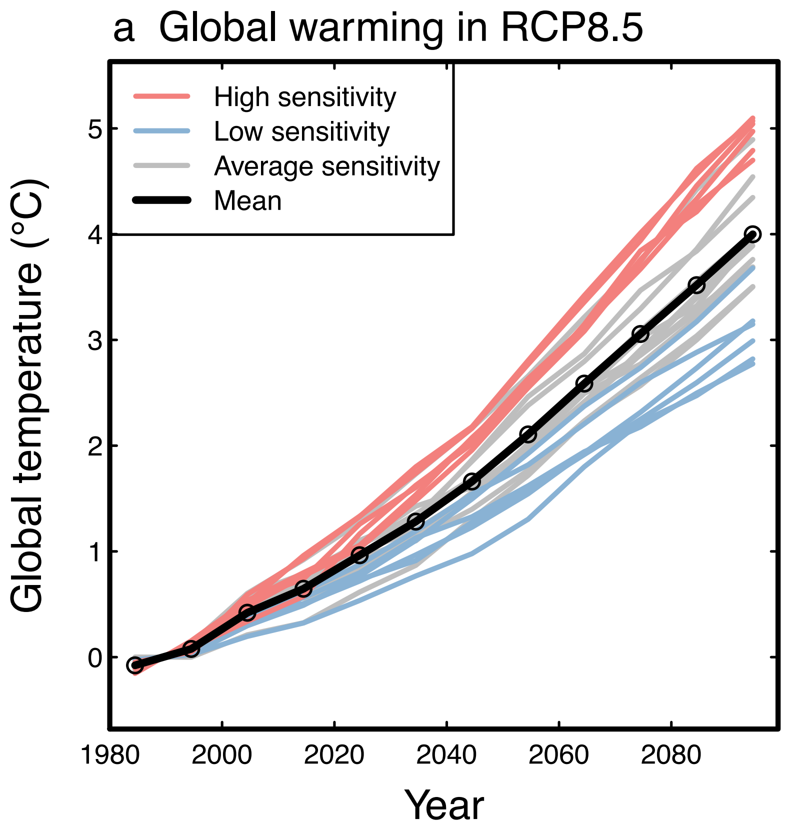
$$0 = F_{2xCO2} - \lambda \Delta T_{2xCO2}$$

- Solve for 2xCO_2 warming:

$$\Delta T_{2xCO2} = F_{2xCO2} / \lambda$$

Climate sensitivity

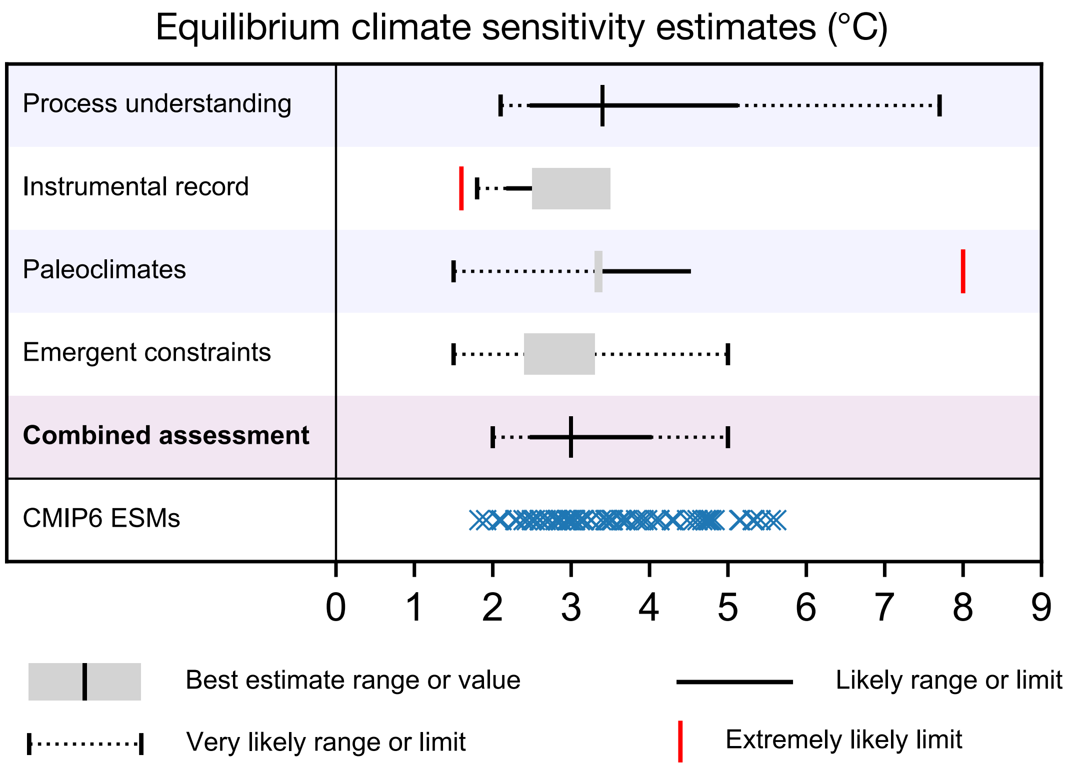
- The value for the climate sensitivity of the Earth is one of the most hotly debated questions in climate science
- Climate models disagree on this number (and thus on temperature projections for a given scenario)
- (RCP8.5 is the high-emissions scenario assessed in the IPCC AR5 – similar to SSP5-8.5)



From Ceppi & Gregory 2020,
Grantham Institute Briefing Note no 11

Climate sensitivity

- Climate sensitivity can be estimated from
  - Climate model experiments with CO_2 forcing
  - The instrumental record
  - Paleoclimate reconstructions
  - Understanding of forcing and feedback processes
  - A combination of these methods
- Regardless of the choice of method, there are large uncertainties
- Much ongoing research on reducing these uncertainties



IPCC AR6 WG1 Technical Summary, Fig. TS.16b

Climate sensitivity

- Why so much uncertainty?
- Recall that

$$\Delta T_{2xCO2} = F_{2xCO2}/\lambda$$

Climate sensitivity

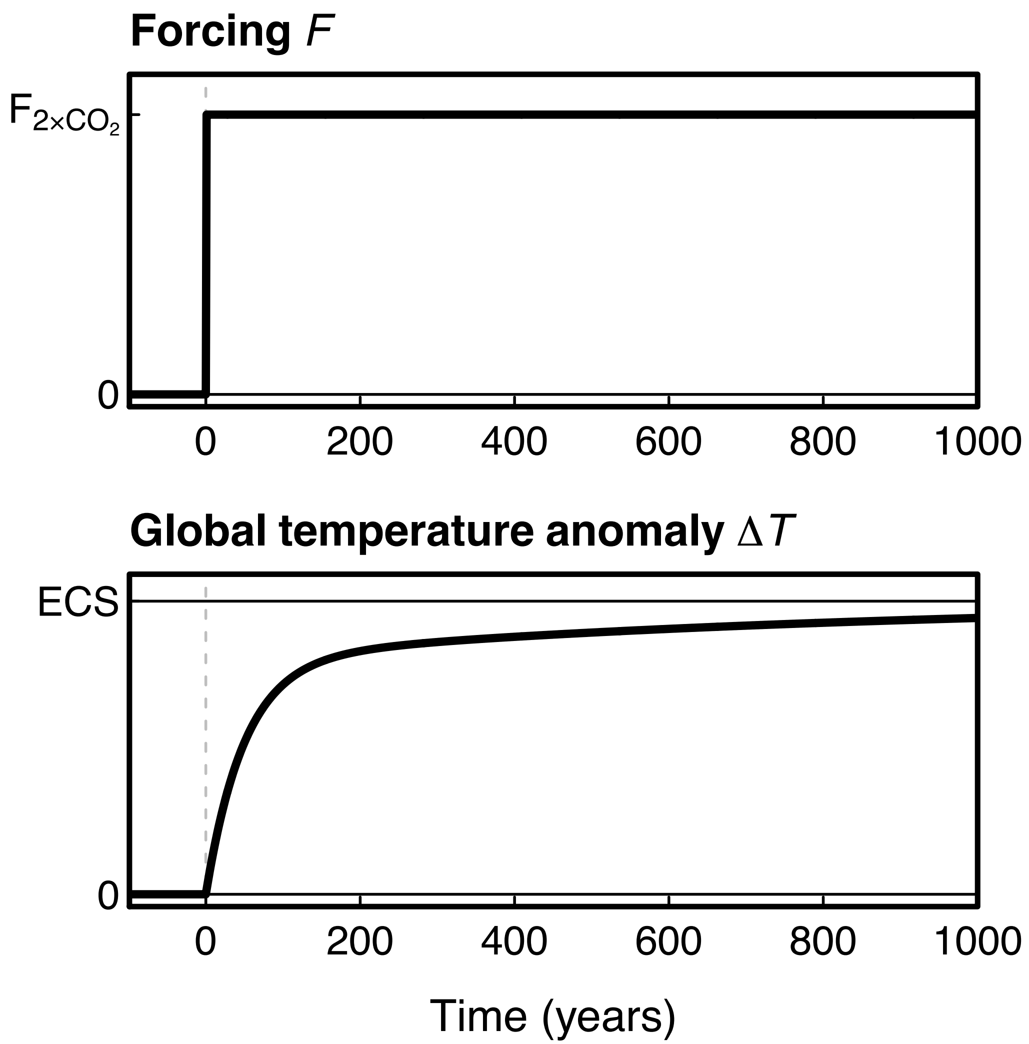
- Why so much uncertainty?
- Recall that

$$\Delta T_{2xCO2} = F_{2xCO2}/\lambda$$

- Most of the uncertainty comes from λ, which measures how efficiently the Earth changes its radiative output to space per unit warming
- λ is known as the **climate feedback** parameter
- In other words: the climate feedbacks determine how much warming is needed to respond to the forcing

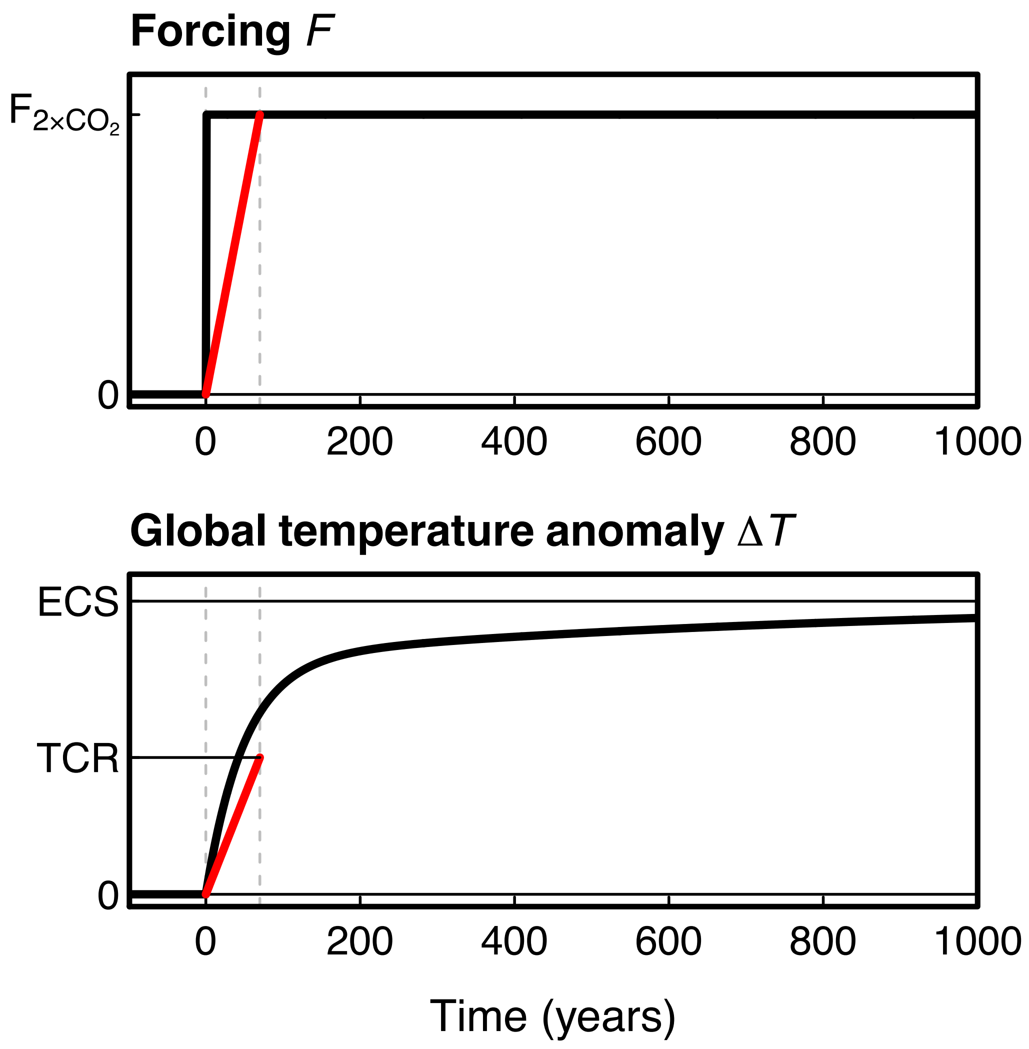
Transient climate response

- Climate sensitivity assumes climate comes back to equilibrium



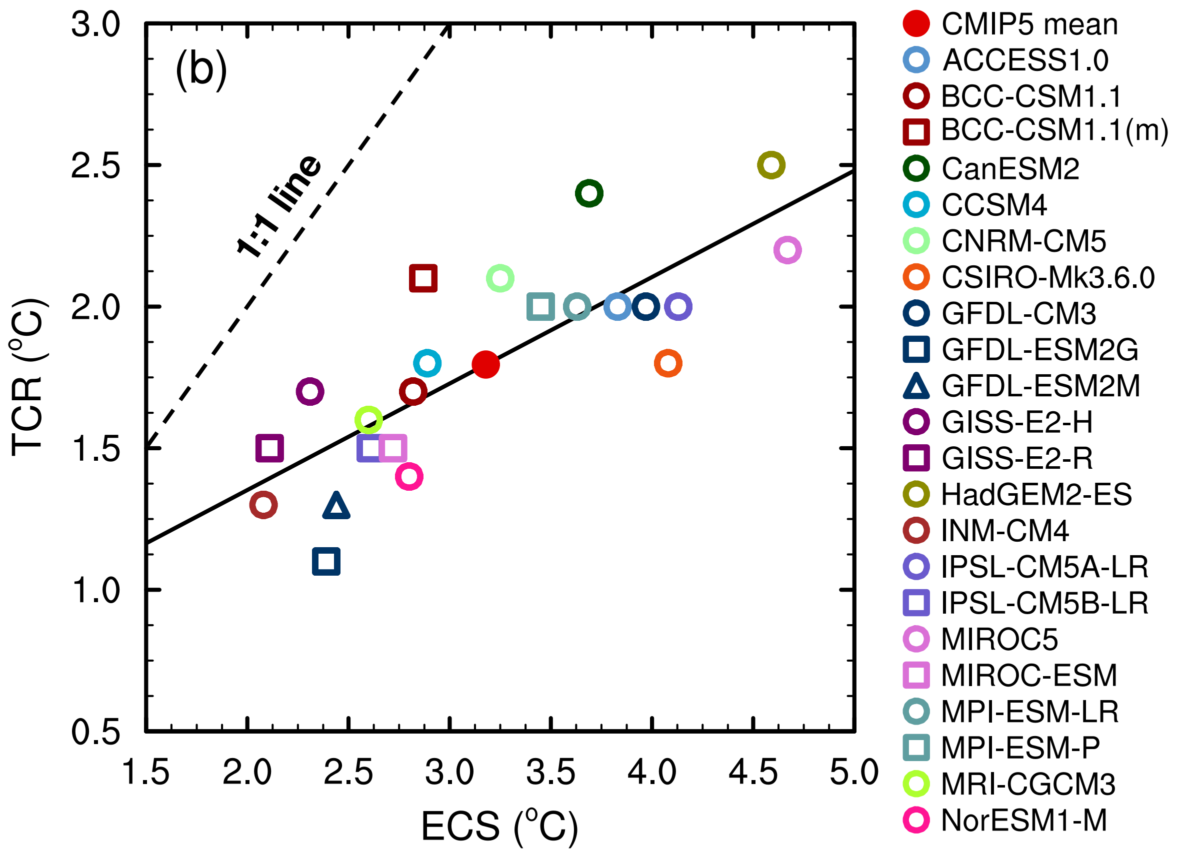
Transient climate response

- Climate sensitivity assumes climate comes back to equilibrium
- The **transient climate response** (TCR) is an alternative metric, more relevant to decadal-scale climate change
- Defined as the amount of global warming at the time of CO_2 doubling, assuming 1% CO_2 increase per year (takes 70 years)



TCR versus climate sensitivity

- TCR and climate sensitivity are not the same, but they are related:
- Climate models with high climate sensitivity have a high TCR, and vice versa



IPCC AR5 WG1 Chapter 8, Fig. 9.42

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Climate feedbacks

- Climate feedbacks are responses to the forcing that amplify or dampen the original forcing
- Example: CO_2 forcing → warming → ice melts → lower albedo → more absorbed solar radiation → warming

a positive feedback

- This is the albedo feedback

Climate feedbacks

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- Example: CO_2 forcing ⟶ warming ⟶ ice melts ⟶ lower albedo ⟶ more absorbed solar radiation ⟶ warming

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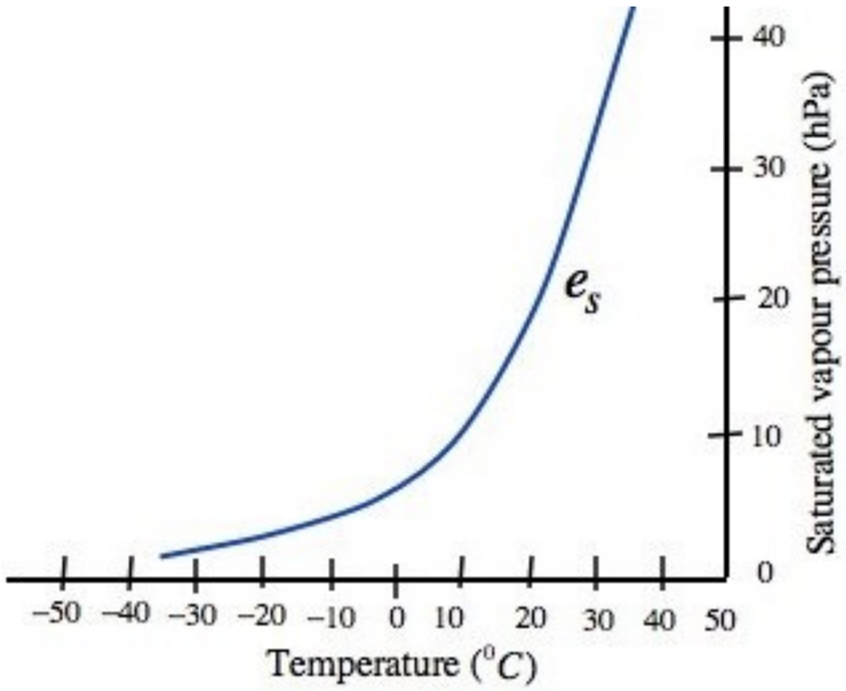
- This is the albedo feedback
- Additional examples:
  - Water vapour feedback
  - Cloud feedback

These are distinct! Water vapour is a **gas**, clouds are **liquid**

Water vapour feedback

- Recall that water vapour is a powerful greenhouse gas
- Water vapour content is tightly controlled by temperature via the **Clausius-Clapeyron relation**
- Maximum vapour content goes up exponentially with temperature
- Relationship predicts 7% increase in water vapour content per degree warming
- Climate models in good agreement on this
- In short: **the water vapour feedback is positive**



Clouds and climate

- Clouds affect the radiative budget in two ways:
  - They reflect solar radiation: a cooling effect
  - They absorb terrestrial radiation (like a greenhouse gas): a warming effect
  - On average, the cooling effect dominates; climate is colder than if clouds didn't exist.
- The question is, how does this radiative effect change when CO_2 increases? I.e., what is the cloud feedback?

Cloud feedback

- The **longwave** greenhouse effect of clouds will increase under global warming. This is a **positive feedback**, associated with the rise of high clouds with warming.
- The **shortwave** cooling by clouds will *probably* decrease, mainly because of decreasing cloud cover in the tropics. This would also be a **positive feedback**.
- However, the shortwave feedback is not robust in models. In fact, **this is where most of the uncertainty in climate sensitivity comes from**.

My research on this…

PNAS Proceedings of the National Academy of Sciences of the United States of America

Observational evidence that cloud feedback amplifies global warming

Paulo Ceppi[a,b,1,2] and Peer Nowack[a,b,c,d,1]

PNAS, July 2021

See also

Carbon Brief: Clouds study finds that low climate sensitivity is ‘extremely unlikely’

Scientific American: Clouds may speed up global warming

Climate feedbacks in IPCC AR5 & AR6

- From IPCC AR5 WG1 SPM:

The net feedback from the combined effect of changes in water vapour, and differences between atmospheric and surface warming is *extremely likely* positive and therefore amplifies changes in climate. The net radiative feedback due to all cloud types combined is *likely* positive. Uncertainty in the sign and magnitude of the cloud feedback is due primarily to continuing uncertainty in the impact of warming on low clouds. {7.2}

- IPCC AR6 WG1 Technical Summary (TS.3.2.2):

"The net effect of changes in clouds in response to global warming is to amplify human-induced warming, that is, the net cloud feedback is positive (*high confidence*). Compared to AR5, major advances in the understanding of cloud processes have increased the level of confidence and decreased the uncertainty range in the cloud feedback by about 50%"

Recap

- Climate sensitivity is a key metric, quantifying how much the world would warm for a doubling of CO_2 concentration
- Climate feedbacks are changes in the system that can amplify or dampen the response to a forcing
- Climate sensitivity is uncertain because feedbacks are not known accurately, especially cloud feedback

Further reading

- Global warming: Understanding the forecast (D. Archer), Chapter 7
- How scientists estimate climate sensitivity (Carbon Brief, June 2018)
- Understanding climate feedbacks (Carbon Brief, October 2015)
- Why clouds are the missing piece in the climate change puzzle (The Conversation, September 2020)
- Climate sensitivity: What is it, and why is it important? (Grantham Institute Briefing note no 11, February 2020)